48 QGIS as a FLOSS Workbench for Multisensor Analysis in Preventive Archaeology: Hybrid Workflows for Infrastructure Design

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UAV-based remote sensing surveys are playing an increasingly important role in preventive archaeology applied to infrastructure design. However, the related workflows are often necessarily hybrid, since data acquisition and pre-processing frequently rely on proprietary hardware and software ecosystems. This paper presents the experience developed by the Struttura Archeologia (Archaeology Unit) of Anas S.p.A. in the integrated use of multispectral, LiDAR and thermal data, proposing QGIS as a FLOSS environment for processing, comparing and interpreting heterogeneous datasets.

The approach does not aim to automate archaeological interpretation, but rather to make the analytical steps that support it more efficient, transparent and reproducible. Through processing tools and parameterized workflows, QGIS can automate the generation of spectral indices, morphometric derivatives and multi-scale visualizations from DTMs, as well as the processing and comparison of thermal data, providing alternative representations for specialist assessment.

The approach distinguishes between automated processing and interpretative optimization: visualization parameters, analysis scales, stretches and display methods are not necessarily fixed in advance, but can be adjusted by the operator according to the characteristics of each dataset.

The workflow therefore automates repetitive operations without constraining those stages in which archaeological expertise and judgement remain essential.

Through selected case studies related to Anas infrastructure projects, the paper explores QGIS as an open-source workbench for multisensor analysis in preventive archaeology, integrating standardized procedures and specialist interpretation within hybrid professional pipelines, while promoting interoperability, reproducibility and methodological reuse.